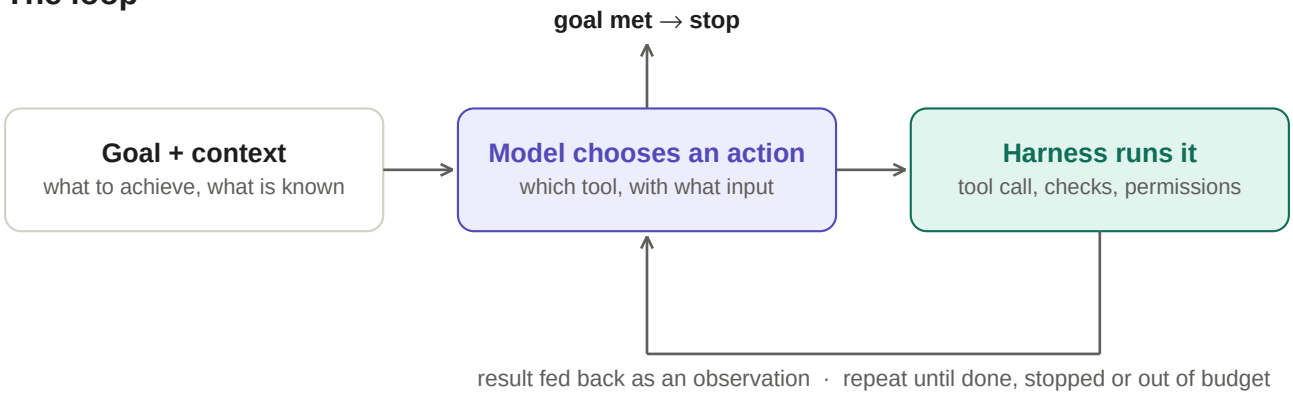
Why an agentic loop exists

An agent harness is the code around a model that runs one cycle until the goal is met. This page covers why that cycle exists, what the harness adds around it, and when a fixed workflow is the better choice.

The loop



Why it exists: the steps cannot be written in advance

1. **The next step depends on the last result.** The model has to see what came back (a file, an error, an image, a test result) before it can know what to do next. A script fixed in advance cannot react to it.
2. **Recovery.** When an action fails or the result is wrong, the agent tries something else (a different input, a different tool, a smaller step) instead of stopping.
3. **Open-ended goals.** "Make this work" has no fixed recipe. The model plans, acts, learns from what happened and re-plans along the way.
4. **Judgement between steps.** Is this result good enough? Is another attempt worth its cost? Which of several options is best? The model makes those calls inside the loop.

What the harness provides around the loop

Tools	the actions the model may take, each with a clear input and output
Permissions and budgets	what needs a human first, and how much time, money or how many attempts it may spend
Stop conditions	goal met, budget spent, or an error the loop cannot fix
Memory and state	what happened so far, so each turn builds on the last
Observation	results turned into something the model can read: outputs, errors, checks

Agentic loop or fixed workflow?

	Fixed workflow	Agentic loop
Who decides the next step	the code, written in advance	the model, from what it observes
Reacts to results	only where a check was coded	every turn
Cost and speed	low, predictable	higher, varies by task
Predictability	high: same input, same steps	lower: the path can differ each run
Best for	steps known in advance	steps that depend on what happens

Rule of thumb: use a fixed workflow when you can write the steps down, and an agentic loop when the right next step depends on what the last one produced. Most real systems mix the two, with a workflow for the known path and a loop where judgement on results is needed.